

News Weekly

Residents describe intense, 5.6 magnitude quake that caused damage, injuries in Mendocino County

A magnitude 5.6 earthquake struck Mendocino County on Wednesday morning, shaking communities and causing scattered damage. The quake occurred shortly after 8 a.m., centered between Ukiah and Willits. More than a dozen aftershocks followed, most registering below 2.5 in magnitude.

The Mendocino County Executive Office reported some injuries but no fatalities. County spokesperson Heather Rose mentioned that hospitals had noted injuries, though specifics were unavailable. Officials planned to gather later Wednesday to assess the situation.

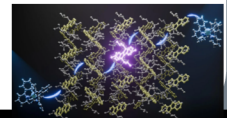
Reports indicated damage such as wine bottles falling off shelves, broken windows, and cracks in buildings. Although quakes of this magnitude usually don't cause extensive damage, residents noted the shaking felt unexpectedly strong. In Willits, north of the epicenter, some residents described it as the most intense quake they had experienced. Surveillance footage from Mariposa Market showed shelves being violently shaken for several seconds. "It was like the store was turned upside down, it was a big mess," said Mark Adley, the market's manager (Source:ABC News)



New solid-state material converts sunlight into higher-energy UV light

Imagine combining two cups of warm water and somehow achieving boiling water. While this is impossible in everyday life, at the quantum level, something similar can happen. Low-energy light particles can merge to form a higher-energy particle. Researchers at Kyushu University have created a solid-state material that efficiently converts visible sunlight into UV light, achieving a photo upconversion efficiency of 1.9%, as reported in a June 23 Nature Communications study.

UV light, though often associated with skin damage, plays a crucial role in air purification, 3D printing, and dental applications. It constitutes about 6% of the sunlight reaching Earth. "We 'add' energy from two visible light photons to make one UV photon, a process called photo upconversion," explains Yoichi Sasaki, a professor at Kyushu University. The process involves triplet-triplet annihilation (TTA), where a donor molecule absorbs light and transfers energy to an acceptor molecule. Proper molecular spacing in solids is crucial for energy transfer without quenching excitons. The researchers used dihydroindenoindene (DHI), modified with alkyl chains, to achieve controlled spacing, resulting in strong luminescence and effective energy transfer. This system efficiently converts visible light into UV light, producing two UV photons for every hundred visible-light photons absorbed in natural sunlight. (source:Science Daily)



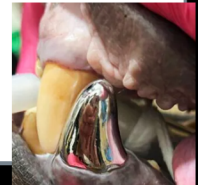
The U.S. can shrug off its last-gasp loss to Türkiye. Now the stakes get up.

The U.S. men's national team faced Türkiye on Thursday without any pressure, having already secured a spot in the World Cup's knockout stage as group winners. Despite a 3-2 defeat, sealed by a last-minute goal from Türkiye's Kaan Ayhan, the result did not affect their standings. The U.S. is set to play Bosnia and Herzegovina in the round of 32 on July 1 in Santa Clara, California. Coach Mauricio Pochettino used the opportunity to rest key players and give seven others their tournament debuts. Christian Pulisic returned from injury and showed his skill, nearly scoring twice. Despite the loss, the team remained optimistic, with midfielder Brendan Aaronson and Sebastian Berhalter praising Pulisic's impact. The U.S. finished the group stage with eight goals, a record for them at a single World Cup. Coach Pochettino expressed satisfaction with the team's performance and readiness for the next round. (Source NBC News)



Alaska brown bear has a new, shiny smile after getting a huge metal crown for a canine tooth

An Alaska brown bear named Tundra at the Lake Superior Zoo in Minnesota recently received a new silver-colored canine tooth, marking a groundbreaking procedure for bears. According to zoo marketing manager Caroline Routley, "He's got a little glint in his smile now." The hour-long procedure was performed by Dr. Grace Brown, a veterinary dentist who previously conducted a root canal on the same tooth. After Tundra reinjured it, they decided on a stronger titanium alloy crown, crafted by Creature Crowns in Idaho. Brown plans to publish a paper on the procedure in the Journal of Veterinary Dentistry, stating, "This is the largest crown ever created in the world." Tundra, who is 6 years old and stands 8 feet tall, has been at the Duluth zoo since he was 3 months old. The procedure required a trained armed response team member in case Tundra awoke, but it went smoothly, and he is now back in his habitat, eating normally. (Source:ABC)

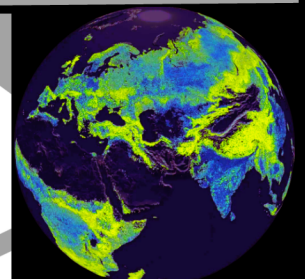


Beneath our feet lies a fungal superhighway stretching 68 quadrillion miles

Beneath our feet lies a vast hidden fungal superhighway that sustains much of life on Earth—and scientists have now mapped it for the first time. The study, published in "Science", focuses on arbuscular mycorrhizal fungi, which form partnerships with most plants. Scientists also released an interactive visualization to explore this underground infrastructure. These maps will help researchers and policymakers identify thriving and threatened fungal networks. Key findings include:

- Global topsoils contain an estimated 110 quadrillion kilometers of arbuscular mycorrhizal fungal network, made up of thread-like hyphae. That distance is almost a billion times the Earth-Sun distance.
- Grasslands contain roughly 40% of Earth's fungal infrastructure, with dense networks in South Sudan, Florida's Everglades, and Tibet.
- AM fungal networks move an estimated 4 billion tons of CO₂e into soils yearly (11% of human-related emissions).

Large agricultural croplands have about 50% lower network densities, potentially reducing a soil's ability to store carbon, cycle nutrients, and



Jokes

What is a dog magician called?

A Labracadra.

Connections



Blueberry Lemonade

Instructions

Combine 1 cup blueberries, agave syrup, and 1 cup water in a medium saucepan.

- Heat over medium heat until the blueberries soften and start to burst, stirring occasionally for about 8–10 minutes.

- Gently mash the blueberries with a fork or masher to release their juices

Stir the fresh lemon juice into the cooled blueberry syrup.

- Add the remaining 3 cups of cold water, tasting to adjust sweetness or tartness as desired. Refrigerate for at least 30–60 minutes for best flavor.

Serve over ice, and garnish with lemon slices, blueberries, or fresh mint for a decorative and flavorful touch

Ingredients (Serves 4–6)

- 1 cup fresh or frozen blueberries
- 1 cup freshly squeezed lemon juice (about 4–6 lemons)
- 2/3 cup agave syrup, or to your taste
- 4 cups cold water
- Ice cubes
- Lemon slices and a few fresh blueberries, for garnish
- Optional: fresh mint leaves for garnish

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